



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Business systems

Academic Year: 2024- 2025 (Odd Semester)

Degree, Semester & Branch: B.Tech.,V & CSBS

Course Code & Title: CW3551 & Data and Information Security

Name of the Faculty member (s): Ms.M. Preethi Ram, AP/CSBS

Innovative Practice Description

Unit / Topic: Unit 3 / Digital Signature

Course Outcome: CO3

Topic Learning Outcome: TLO22

Activity Chosen: Learning by Teaching

Justification:

Learning by teaching is a method of teaching in which students are made to learn material and prepare lessons to teach it to the other students. There is a strong emphasis on acquisition of life skills along with the subject matter. Digital Signature is a fundamental concept in data and information security and have wide-ranging applications. As students take on the role of teachers, they engage in discussions, answer questions, and share insights with their peers. This peer-to-peer interaction not only reinforces their own understanding but also allows for diverse perspectives and approaches to problem-solving, enriching the overall learning experience.

Time Allotted for the Activity: 35 minutes

Details of the Implementation:

- Before the class, the teacher, inquired about the students' interest in teaching the concept of digital Signature few students expressed their interest.
- Based on their interest Jana Jophin and Guru Vidhya were ready to present their ideas to the students.
- The adequately time was given to the students to prepare for their teaching role so the topic was given a day before the class.
- Students presented their allocated topics to their classmates
- To keep the class interested, students utilized a variety of instructional strategies. Jana Jophin employed real-world examples of digital signature.
- To promote involvement, students were allowed to ask questions to their peers during the presentation. Because of this, students were able to have conversations and share their perspectives about topic which very important in programming.

CO – PO / PSO mapping:

CO	PO1	PO2	PO4	PO9	PSO1
CO 3	3	3	2	1	3

(1 – Low 2 – Moderate 3 High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO4	PO9	PO 10	PSO1
	3	2	1	1	1	3
Justification for correlation	Students can apply specific knowledge from fields like computer science, electrical engineering, or information security to understand digital signature protocols.	Students can identify complex issues such as vulnerabilities in digital signature algorithms or the impact of digital signatures on data security.	Students learn to critically evaluate the effectiveness of different digital signature methods based on their experimental results.	Students can work in teams to prepare and present on various aspects of digital signatures, such as the technical mechanisms, real-world applications, and security implications.	Students can be able to prepare to teach their peers about digital signatures, they must communicate complex concepts clearly and concisely, enhancing their ability to explain technical material.	Students can explore how digital signatures are integral to cyber security protocols, protecting data integrity and authenticity in digital communications.

Images / Screenshot of the practice



Fig.1. Jana Jophin presented the topic to the students



Fig.2. Guru Vidhya shared her ideas to her classmates

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- All the students were eagerly listened the class. Students felt they are able to understand the topic.
 - The students were able to remember key details and concepts more easily.
 - The students express that role of a teacher boost their confidence in knowledge.
 - Students highlighted that this practice helped them improve their communication skills, such as clarity, organization, and the ability to present complex ideas in a simplified manner.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- Students actively engaging in teaching their peers gain a more profound understanding about digital signature.
 - Actively participating in the teaching process helps students remember key details and concepts better compared to passive learning methods
 - This activity helped the student to improve communication skills.
 - Students spent their time in self-learning.
 - This activity encouraged the students to share their knowledge with others.

- ❖ ***Challenges faced in implementation:***

- Effectively motivate students who are not participating in the activity due to shyness, fear of public speaking, or lack of confidence. Motivate the students to overcome their shyness and engage in active learning.
 - Implementing Learning by Teaching required additional time compared to traditional teaching methods. Planning and executing interactive teaching sessions was challenging.

References:

1. <https://effectiviology.com/protege-effect-learn-by-teaching>
2. https://www.researchgate.net/publication/351905566_Impact_of_Seminars_on_Student_Soft_Skills_Development

Signature of Faculty Member

HOD